

Surveillance in India and Its Privacy Challenges in the Digital Age: A Legal and Constitutional Analysis

Growing surveillance framework in India, analyzing its impact on individual privacy, legal safeguards, constitutional rights, and the challenges of emerging digital technologies in the absence of robust data protection laws.

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Abstract

In the digital era, balancing national security with individual privacy rights is increasingly complex, particularly in India, where the expansion of mass surveillance technologies raises significant concerns. The Indian government's new mass surveillance initiatives pose a serious threat to the right to privacy. With large-scale interception of communications, keyword tracking, and easy access to users' data, these systems raise alarming concerns—especially since India's existing privacy safeguards are already inadequate for even targeted surveillance. When expanded to mass surveillance, the risks become far greater, potentially undermining the foundations of democracy. The widespread collection of vast amounts of personal data, enabled by advanced technology, leaves this information vulnerable and poorly protected. This paper explores the relationship between privacy rights and government surveillance in India, shedding light on the dangers of unchecked mass surveillance. By examining India's privacy jurisprudence and the principles guiding it, the paper assesses whether current legal protections can withstand the challenges posed by mass surveillance. It also highlights the potential harm that could arise if such surveillance continues without proper oversight. This research examines the tension between enhancing national security through advanced surveillance measures and protecting citizens' privacy rights. By analyzing legal frameworks such as the Information Technology Act and the Digital Personal Data Protection Act, alongside landmark judicial rulings like *K.S. Puttaswamy vs. Union of India* (2017), the study highlights how India's legal system attempts to navigate this balance. Surveillance technology is crucial in strengthening security, enhancing public safety, improving resource management, and supporting emergency response efforts. However, it also raises serious concerns about privacy violations, potential misuse, discrimination, and a lack of transparency and accountability. This paper underscores the need for a balanced approach that leverages surveillance's benefits while protecting individual rights. It highlights the importance of strong legislation, public awareness, technological safeguards, and collaborative efforts to create responsible surveillance frameworks. Additionally, the paper examines how surveillance practices vary worldwide, considering cultural and societal factors and the complexities of cross-border data sharing.

Ultimately, it calls for a thoughtful and ethical approach to the future of surveillance technology that prioritizes security and personal privacy while safeguarding social values and democratic principles.

Keywords: National security, Privacy rights, Mass Surveillance, Personal data, Principles.

I. INTRODUCTION

In the 21st century, surveillance has become a crucial yet contentious issue, significantly impacted by rapid technological advancements and an expanding digital landscape. In India, the telecommunication and internet usage surge has heightened concerns over the interception and monitoring of personal data. When properly regulated, surveillance can be vital for ensuring national security and preventing criminal activities. However, the intersection of surveillance with individual privacy has raised serious legal and ethical questions. India's legal framework for surveillance is primarily governed by two key statutes: the Indian Telegraph Act of 1885 and the Information Technology Act of 2000. The former deals with the interception of telephone communications, while the latter addresses electronic communications. These laws, along with the Information Technology (Procedure and Safeguards for Interception, Monitoring and Decryption of Information) Rules, 2009, and the more recent Digital Personal Data Protection Act of 2023, outline the parameters for lawful surveillance and data protection¹. The Pegasus scandal, which exposed unauthorized tracking of prominent individuals' data, has intensified scrutiny of India's surveillance practices. The Supreme Court of India has emphasized privacy as a fundamental right, highlighting the need for robust legal protections against unlawful surveillance.

Despite this, recent developments indicate that the government's approach to surveillance remains opaque and lacks accountability, with significant concerns about the concentration of power within the executive branch and the absence of independent oversight. Technological advancements have further complicated the surveillance landscape, with state agencies employing sophisticated tools to monitor and aggregate citizen data. This has led to a chilling effect on public participation and

¹ Daragh Murray, Pete Fussey, Kuda Hove, Wairagala Wakabi, Paul Kimumwe, Otto Saki, Amy Stevens, The Chilling Effects of Surveillance and Human Rights: Insights from Qualitative Research in Uganda and Zimbabwe, *Journal of Human Rights Practice*, Volume 16, Issue 1, February 2024, Pages 397–412.

dissent, with civil society members reporting increased self-censorship and fear of being monitored.² Cases involving journalists and activists reveal the growing tension between national security objectives and preserving privacy and free expression. The ongoing debate about surveillance in India underscores the need for a balanced approach that protects individual rights while addressing security concerns. Reforms should enhance transparency, establish independent oversight mechanisms, and ensure surveillance practices adhere to legality, necessity, and proportionality principles. As India navigates these challenges, it must balance safeguarding national security and upholding democratic values. As the surveillance landscape in India rapidly evolves with technological advancements and legal frameworks shifts, several critical research questions emerge. These include: How effectively do current Indian laws, such as the Indian Telegraph Act and the Information Technology Act, address modern surveillance challenges and safeguard individual privacy? What are the limitations of existing statutory and judicial mechanisms in regulating state surveillance and ensuring accountability? How do emerging technologies and their applications impact the balance between national security and personal privacy? The primary problem under investigation is the inadequacy of current legal and oversight frameworks to manage the complexities of contemporary surveillance practices while protecting fundamental rights. This research aims to evaluate the effectiveness of India's surveillance laws and oversight mechanisms, analyze the implications of technological advancements on privacy and public participation, and propose actionable reforms. The objectives are to identify gaps in the current legal framework, assess the impact of surveillance technologies on civil liberties, and recommend strategies for aligning surveillance practices with international human rights standards and constitutional principles.

II. SURVEILLANCE IN INDIA

Surveillance has evolved significantly with advancements in technology. From traditional monitoring forms to sophisticated systems involving artificial intelligence and extensive data collection, surveillance plays a crucial role in security, law enforcement, and commercial activities³. In India, the growth of surveillance technologies has led to the development of various mechanisms by state and private actors.

The concept of surveillance has evolved over time, yet there is still no clear, universally accepted definition—especially in modern technology. The rise of new monitoring methods, such as sousveillance (where individuals watch those in power), has further complicated the discussion around what surveillance truly means today. Ross Bellaby offers a broad definition, describing surveillance as a range of activities beyond traditional CCTV monitoring. It now includes digital tracking methods like "dataveillance" and data mining, where an individual's identity, movements, associations, and activities are constantly observed and analyzed under a watchful eye.

Types of Surveillance:

Surveillance can be broadly categorized based on who conducts it and who is affected. This classification helps in understanding the scope and impact of surveillance practices.

Based on the Actor:

- **State Surveillance** - State surveillance involves government agencies or law enforcement bodies monitoring individuals or groups. This type of surveillance is typically aimed at ensuring national security, maintaining public order, and enforcing laws. State surveillance has become more sophisticated in India, and advanced technologies, including data collection and monitoring systems, have been integrated.
- **Private Actor Surveillance** - Private entities, such as corporations, businesses, or individuals, also engage in surveillance for various reasons. Corporations may monitor employee activities or track consumer behavior for market research. Businesses might use surveillance to protect their assets or prevent fraud. In recent years, the growth of the digital economy has amplified the role of private surveillance, particularly in data collection and analysis.

Based on the Affected Person:

- **Mass Surveillance** - Mass surveillance refers to monitoring a large population segment without targeting specific individuals. This can include public cameras, internet monitoring systems, and data analytics tools. Mass surveillance aims to collect broad data sets that can be analyzed for patterns indicative of criminal or suspicious behavior.
- **Targeted Surveillance** - Targeted surveillance is directed at specific individuals or groups. This form of surveillance is often used when there is a particular concern or suspicion about a person or entity. It involves more focused data collection and analysis, such as wiretapping, email monitoring, and detailed background checks.
- **Lateral Surveillance** - Lateral surveillance occurs when individuals monitor their peers or community members. With the rise of social media and digital communication, this form of surveillance has become more prevalent. During crises like the COVID-19 pandemic, community-driven surveillance became common, where individuals reported on each other's health status⁴.

² Eck, K., & Hatz, S. (2020). State surveillance and the COVID-19 crisis. *Journal of Human Rights*, 19(5), 603–612.

³ Costanza, S.E. (2018) 'Surveillance', in A.J. Treviño (ed.) *The Cambridge Handbook of Social Problems*. Cambridge: Cambridge University Press, pp. 95–108.

⁴ Arun C., 'Paper-thin Safeguards and Mass Surveillance in India' [2014] 26 NLSIR 105

Mechanisms of State Surveillance in India:

India has implemented several sophisticated mechanisms to enhance its surveillance capabilities. These mechanisms gather real-time data, monitor communications, and analyze online activities.

1. National Intelligence Grid (NATGRID)

The NATGRID is a critical component of India's surveillance infrastructure. It provides law enforcement and intelligence agencies with real-time information from various sources, including income tax records, banking data, insurance details, and railway records. The grid aggregates data across 21 categories, allowing for comprehensive analysis and facilitating timely responses to potential threats.

2. Central Monitoring System (CMS)

The CMS is a centralized system that enables direct monitoring of telecommunications. It bypasses traditional telecom providers to intercept and record phone calls, text messages, and social media communications. The CMS enhances the ability of security agencies to monitor communications without relying on intermediary service providers⁵.

3. Network Traffic Analysis (NETRA)

NETRA focuses on monitoring internet communications and online activities. Using keywords and filters, it can analyze text-based messages, including emails and social media posts. NETRA can identify suspicious content even within encrypted messages, making it a crucial tool for cyber surveillance.

4. Automated Facial Recognition System (AFRS)

The AFRS, utilized primarily by the Delhi Police, employs an extensive database of facial images to identify individuals. By matching new photos with the database, the system can determine the identity of individuals, which is particularly useful in public spaces and for tracking suspects.⁶

5. Crime and Criminal Tracking Network System (CCTNS)

CCTNS is designed to streamline the collection, storage, and sharing of data related to crime and criminals. It enables the transfer of information between police stations and state headquarters, enhancing coordination and efficiency in law enforcement operations.

III. PRIVACY AND SURVEILLANCE

India's recognition of the right to privacy as a fundamental right has been a long and evolving journey. It traces its roots back to the Kharak Singh case of 1962, where the Supreme Court first considered privacy in a legal context. However, at the time, privacy was not explicitly acknowledged as a constitutionally protected right. It wasn't until the landmark Justice K.S. Puttaswamy judgment in 2017 that the Supreme Court firmly established privacy as a fundamental right under Article 21 of the Indian Constitution. This ruling marked a significant turning point, affirming that privacy is an essential aspect of personal liberty and dignity. The journey from Kharak Singh to Puttaswamy reflects India's growing recognition of the need to protect individuals from excessive state surveillance and intrusion into their personal lives. The bench considered the right to privacy in its full context for the first time in the Kharak Singh case, although it was not recognized as a right guaranteed by the Constitution. The bench had connected the effect of law enforcement agencies' surveillance mechanisms on the petitioner's right to privacy.⁷ However, the bench in the PUCL case pronounced the judgment in support of Justice Subba Rao's dissenting opinion in the Kharak Singh case, which led to the expansion of the scope of Article 21 to include the "right of an individual to be free from restrictions or encroachments on his person."

The ongoing debate between state surveillance and individual privacy has been significantly shaped by judicial interventions in India. Courts have played a crucial role in defining the limits of lawful surveillance while also ensuring that citizens' rights remain protected. Judicial rulings have influenced not just what types of surveillance are permissible, but also how and under what conditions they can be carried out. One of the most transformative rulings in this context was the K.S. Puttaswamy v. Union of India case. This historic judgment by a nine-judge bench of the Supreme Court redefined India's privacy laws. It overturned previous rulings that had denied the existence of a constitutional right to privacy, firmly establishing that privacy is an inherent part

⁵ Chaudhari N., Smitha Krishna Prasad, 'Carpenter v. United States: State Surveillance and Citizen Privacy' [2019] 13 NALSAR Stud L Rev 129

⁶ Karl Ricanek Jr and Chris Boehnen, 'Facial Analytics: From Big Data to Law Enforcement' (2012) 45(9) Computer 95, 95

⁷ Xiu J., 'The Roles of the Judiciary in Examining and Supervising the Changing Laws of Electronic Surveillance' [2003]

of the right to life and personal liberty under Article 21 of the Constitution. The Puttaswamy verdict laid down key principles that now guide privacy rights and data protection in India. It clarified what qualifies as private, why privacy is fundamental to human dignity, and how it must be protected against state overreach. This judgment has since influenced policies and legal frameworks, reinforcing that any intrusion into privacy must be justified, necessary, and proportionate to the objective it seeks to achieve.

- **Privacy as a Fundamental Right:** The judgment ruled that privacy is a fundamental right, ensuring it receives the highest legal protection against arbitrary state action.

- **Three-Part Test for Privacy Invasions:** It established a three-part test for any government action seeking to breach fundamental rights of privacy: legality (a law exists); necessity (a legitimate state need); and proportionality (the least restrictive means to achieving the objective).

- **Data Protection Principles:** The Court also emphasized protecting your data. It suggested that it is essential to create robust laws to protect personal data from unlawful use.

The K.S. Puttaswamy judgment has become the cornerstone for evaluating laws and policies related to surveillance, data protection, and privacy in India. It has influenced numerous legal decisions that seek to balance state authority with individual rights.

One such case was *People's Union for Civil Liberties (PUCL) v. Union of India*,⁸ where the Supreme Court laid down guidelines to prevent arbitrary phone tapping under the Indian Telegraph Act, 1885. The Court emphasized that wiretapping is an intrusion into a person's privacy and can only be permitted if it follows a legally established procedure. This case set the groundwork for further defining privacy rights, ultimately paving the way for the more detailed privacy protections outlined in Puttaswamy.

Another significant case was *Justice K.S. Puttaswamy (Retd) v. Union of India*, commonly known as the Aadhaar Judgment. Following the 2017 privacy ruling, the Supreme Court assessed the constitutionality of the Aadhaar project. While the Court upheld the use of Aadhaar for government welfare schemes and PAN linkage, it struck down provisions that mandated Aadhaar for mobile connections and bank accounts, ruling that such requirements violated the right to privacy. The judgment reinforced the idea that while the state can collect biometric data for legitimate purposes, such collection must be necessary, proportionate, and not infringe on privacy arbitrarily.

The *Navtej Singh Johar v. Union of India* case, though not directly related to surveillance, further expanded the scope of privacy rights. This landmark ruling decriminalized consensual same-sex relationships, striking down Section 377 of the Indian Penal Code. The Supreme Court held that privacy includes "the right to be let alone", emphasizing that personal and intimate choices—especially regarding sexual orientation—must remain free from outside interference. By recognizing decisional privacy, the Court extended privacy protections beyond data and surveillance, strengthening its status as a fundamental right.

Collectively, these judgments have shaped India's privacy jurisprudence, ensuring that while the state may conduct surveillance, it must do so under strict legal safeguards. These rulings reflect an evolving awareness of how technology can threaten personal freedoms, while also recognizing the legitimate security needs of the state. By establishing clear legal principles, these cases provide a constitutional foundation for future legislation and policies governing privacy and surveillance in India.

Threat to Privacy Due to Surveillance

State surveillance poses several significant threats to individual privacy, fundamentally altering how personal information is handled and protected. One primary concern is the mass collection of data. Governments may accumulate extensive information on individuals, from their communications and internet activities to detailed personal records. This broad data collection can lead to serious privacy violations, as individuals' private lives are scrutinized without explicit consent.

Another issue is the chilling effect that surveillance can have on free expression.⁹ When people know they are being monitored, they may self-censor or avoid engaging in certain activities due to fear of repercussions. This suppression of free speech and participation can undermine democratic processes and stifle healthy public discourse. Moreover, the potential for abuse of power is a significant risk. Authorities can misuse surveillance tools and the data they generate for political, personal, or discriminatory purposes. Such misuse can result in harassment, oppression, or targeted actions against individuals or specific groups, raising severe ethical and legal concerns. The security of collected data also presents a critical threat. Sensitive information gathered through surveillance is vulnerable to breaches, hacking, or unauthorized access. If this data is compromised, it can lead to significant privacy violations and security risks for those affected.¹⁰ A related concern is the lack of oversight and accountability in many surveillance programs. Surveillance practices can become invasive and unchecked without adequate checks and balances, eroding public trust in government institutions and diminishing the rule of law.

Furthermore, the normalization of surveillance can gradually erode societal norms surrounding privacy. As monitoring becomes more common, the boundaries of acceptable surveillance can expand, leading to a society where privacy is increasingly compromised and individual freedoms are diminished. Surveillance also poses risks of false positives and misidentification. Automated systems like facial recognition technology can sometimes incorrectly identify individuals or flag innocent people as suspects. This can lead to wrongful actions, stigmatization, and an overall sense of insecurity.

Finally, the erosion of anonymity is a significant concern. Surveillance reduces the ability to remain anonymous, essential for privacy and freedom of expression. This impact extends to online and offline activities, making individuals more vulnerable to

⁸ Ramachandran C., 'PUCL v. Union of India Revisited: Why India's Surveillance Law Must Be Redesigned for the Digital Age' [2014]

⁹ Schauer F., 'Fear, Risk and the First Amendment: Unraveling the "Chilling Effect"' [1978] 58 BU L Rev 693

¹⁰ Addison Litton, "The State of Surveillance in India: The Central Monitoring System's Chilling Effect on Self-Expression", Washington University Law Review, 2015

https://openscholarship.wustl.edu/cgi/viewcontent.cgi?article=1556&context=law_globalstudies, at 816

scrutiny and control. Addressing these threats requires a careful balance between security and privacy, with robust oversight mechanisms to ensure surveillance practices do not undermine fundamental rights and freedoms.

Overreach of Surveillance Powers

India's approach to surveillance for national security purposes, while crucial for protecting state interests, has faced substantial criticism over overreach and misuse of power. The current legal framework is often criticized for its lack of transparency and independent oversight, which, compounded by the absence of a dedicated privacy law, raises fears of potential abuse. Surveillance systems like the Central Monitoring System (CMS) and NATGRID, designed for national security, are feared to facilitate mass surveillance and erode privacy without adequate judicial oversight or probable cause. Additionally, surveillance practices, especially during civil unrest or in sensitive areas, have been seen as infringing on civil liberties, including freedom of speech. The misuse of sedition laws and the Unlawful Activities Prevention Act (UAPA), in conjunction with surveillance data, has been widely reported, highlighting the risks to human rights activists and dissenters. The judiciary has occasionally intervened to address these issues, as evidenced by the Supreme Court's ruling in *K S Puttaswamy*, which underscored the importance of privacy and set boundaries for surveillance practices. This ongoing debate reflects the complex balancing act between ensuring national security and upholding democratic freedoms. It underscores the need for continued judicial scrutiny and legislative reform to develop a more nuanced approach that safeguards security and individual privacy rights.

National Security vs. Privacy Rights

The debate between national security and individual privacy underscores a fundamental dilemma in democracies: whether enhanced surveillance or safeguarding personal privacy should be prioritized. Proponents of increased surveillance argue that it is essential for addressing threats such as terrorism, cybercrime, and external aggression. In countries like India, where geopolitical tensions and internal terrorism are significant concerns, robust surveillance is crucial for thwarting threats, monitoring cyber activities, and defending national sovereignty. They argue that surveillance helps preempt attacks and manage cybercrimes, which have become increasingly sophisticated in the digital age. However, critics assert that prioritizing surveillance over privacy risks substantial misuse and abuse of power. Without stringent safeguards, collected data could be exploited for political repression or other rights violations, undermining democracy. They argue that mass surveillance erodes the foundational principles of openness and accountability in a democracy, chilling free speech and stifling dissent. The fear is that individuals may self-censor or alter their behavior to avoid scrutiny, which not only impinges on democratic freedoms but also hinders innovation and social progress. Thus, the discourse reveals a complex interplay between ensuring security and preserving individual rights, highlighting the need for a nuanced approach that balances both concerns.

Safeguards to the Issue

Legal reforms are imperative to align outdated laws with contemporary privacy standards and technological advancements to address the evolving challenges in privacy and surveillance. The Indian Telegraph Act of 1885 and the Information Technology Act of 2000 need significant updates to incorporate explicit privacy safeguards. Strengthening oversight mechanisms is crucial, including enhancing judicial review, establishing independent bodies or tribunals to grant surveillance warrants, and ensuring such measures adhere to the rule of law. Parliamentary oversight should also be mandated, with regular reporting on surveillance powers and activities to ensure transparency and accountability. This approach will help maintain a democratic balance between national security and individual rights, ensuring that surveillance practices are justified and proportional.

Looking to the future, India must address the rapid advancements in technology that impact privacy. Artificial intelligence (AI) and facial recognition technologies offer significant national security benefits but pose privacy and discrimination risks. The proliferation of Internet of Things (IoT) devices further complicates privacy concerns by collecting susceptible data. To mitigate these risks, new legislation must introduce robust data protection laws, similar to the European Union's General Data Protection Regulation (GDPR), outlining individuals' rights and state obligations. Technological solutions such as end-to-end encryption and data anonymization can help safeguard privacy while allowing lawful surveillance. These measures and solid oversight mechanisms are essential for balancing national security needs with protecting individual privacy rights.

IV. CONCLUSIONS AND SUGGESTIONS

The 21st century has witnessed a profound transformation like surveillance, driven by rapid technological advancements. This evolution has significantly expanded governments' capacity to monitor and intrude into personal spaces, raising critical concerns about privacy and individual rights. In India, this complex issue is shaped by a range of statutes and regulations, including the Indian Telegraph Act of 1885, the Information Technology Act of 2000, and more recent frameworks like the Digital Personal Data Protection Act of 2023. These laws and licensing agreements, such as the Unified License Agreement, create a nuanced landscape of state surveillance, which can be categorized into targeted and mass surveillance powers. Legal and judicial frameworks offer protections against arbitrary surveillance. Statutory barriers are embedded within these laws, delineating specific conditions under which surveillance can be conducted.

In contrast, judicial barriers have evolved through landmark rulings, such as the Supreme Court's decision in *K.S. Puttaswamy v. Union of India*, which enshrined privacy as a fundamental right. These barriers aim to safeguard citizens from the overreach of surveillance powers. Still, the real-world application often reveals gaps, as evidenced by instances where illegally obtained evidence has been used to undermine judicial proceedings.

In examining India's surveillance apparatus, the reports from the Justice Shah and Justice Srikrishna Committees highlight the balancing act between state security and individual privacy. Despite the progressive stance of the Supreme Court, there is ongoing concern that recent legislative efforts, such as the Personal Data Protection Bill, may still grant excessive leeway to government agencies. The broad exemption clauses proposed in the Bill raise alarm about potential abuses, necessitating urgent legal reform to align with international human rights standards.

The advent of emerging technologies like artificial intelligence, facial recognition, and the Internet of Things introduces both opportunities and challenges for surveillance and privacy. While these technologies can enhance national security, they pose significant personal privacy and civil liberties risks. Legislative and policy reforms must address these issues by implementing robust data protection laws, improving oversight mechanisms, and ensuring that technological advancements do not undermine fundamental rights. The path forward for India involves a delicate balance between enhancing national security and preserving individual privacy. Legal reforms must be comprehensive, technology-specific solutions should be developed, and oversight mechanisms must be strengthened to prevent abuse. Civil society's role in advocating for these reforms remains crucial, as evidenced by successful campaigns against intrusive measures like the Aarogya Setu app mandate.

In conclusion, India's surveillance and privacy landscape reflects the global struggle to balance security needs with personal freedoms. As the country navigates rapid technological changes and security challenges, it is imperative to uphold democratic principles, ensure robust legal protections, and foster transparent governance. The future of India's surveillance state will hinge on maintaining this balance, safeguarding individual rights, and preserving the democratic fabric of the nation.

